

PRACTICAL INTRODUCTION TO RESEARCH

Tuesday & Thursday | PMA 15.201 | 2:00p-3:15p

Instructor

Prof. Steven Finkelstein

(he/him/his)

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Help Sessions (HS):

M 1:05-2p: PMA 16.316

Th 315p: PMA 15.201

Teaching Assistants

TA: Natalia Villanueva

nataliavillanueva@utexas.edu

HS: T 1-2p: PMA 16.216

W 1130-1230: PMA 15.201

Reading & Resources:

All readings are free and publicly available, and available on Canvas.

- **"Beginners Guide to Working with Astronomical Data"** by Marcus Possel
- **"Astronomy"**, by Franknoi, Morrison and Wolff – an introductory astronomy textbook, covering a wide variety of material and is intended as a general resource (other introductory textbooks are fine too).

What is this course about?

This course will prepare you both for participating in, and ultimately leading, astronomy research, as well as to be able to use computer coding for your upper-division classes. The class is oriented around practical exercises including analysis of imaging data, visualization, programming exercises with Python, statistical analyses, and training for papers and oral presentations. The class culminates in a final research project which you will work on in a small group.

NOTE: This course is restricted to science and engineering majors. It is not intended to be taken as a general research methods class and it not likely to be useful to those not majoring in astronomy and/or interested in conducting astronomy research. problems.

Course Learning Objectives: By fully engaging in this course, you will...

- **Describe** what research is, and how astronomers use the scientific method to uncover truths about the universe.
- **Apply** knowledge of command-line syntax to navigate filesystems, manipulate data, and launch programs
- **Demonstrate** proficiency with the Python coding language to read and analyze data, and make informative and aesthetically pleasing plots
- **Employ** the NASA ADS system to find journal articles, and **practice** the ability to efficiently read them
- **Summarize** key results and convey information accurately and concisely in both a poster and short talk format
- **Manipulate** astronomical images with the SAOImage ds9 software
- **Develop** code to complete a small, self-contained research project from beginning to end, presenting the results in a talk

Each class period will have learning objectives linked to one or more of these course goals

Course Webpage:

<https://utexas.instructure.com/courses/1422452>

Prerequisites:

Mathematics 305G or the equivalent. An introductory astronomy course (AST307) is **highly recommended**. Prior computing experience is recommended but not required. Be prepared to carry out simple calculations and think independently!

Computing:

You will need a variety of Python packages and software installed on your computer to do the class assignments. All necessary software will be available on the classroom computers. If you wish to use your laptop, you are responsible for determining how to install the necessary tools.

What is expected of me in this class?

- This is an active learning class, meaning that you must attend class and be prepared to **participate**, work collaboratively and share your ideas.
- Complete all assigned modules and assignments on time. Make sure to take time outside of class to complete your in-class activities, and come get help if you have questions!
- Don't procrastinate on the homework or the research project! Be prepared to spend 2-3 hours outside of class for each hour of class time (**e.g., 5-8 hours per week**).
- Take advantage of your teaching team, we are here to help you!

What happens in lecture?

- Not much lecturing! I have designed this class from the ground up to be very interactive, focusing on your attainment of the course learning outcomes. You will be exposed to the relevant content through a short lecture, then it all be time to practice!
- You will only learn if you participate, thus attendance and participation are **required**!
- A typical class day will be composed of the following:
 - Science tidbit of the day.
 - Small (~20-30 minute) presentation of new content
 - An activity where you will use your computer, working with a partner (on their own computer) to practice the material just discussed.

My Commitment to all Students Succeeding

The University of Texas at Austin serves a diverse student body, and it is my intent that all students are able to succeed in this course, regardless of their background. We are different in many ways, and this is fundamental to building and maintaining an inclusive campus community. Astronomy belongs to all people, independent of race, religion, gender, gender identity, gender expression, or sexual orientation. Incidents of discrimination, assault, harassment, threats, intimidation, profiling, or coercion based on membership or perceived membership will not be tolerated. Show each other respect no matter perceived knowledge or performance in this class, or any other.

I pledge to treat all of you with respect, and I ask that you do the same in return to myself and to your classmates. In the classroom, I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use. I ask you to support one another as you talk about differences in your personal experiences whether relevant to the class or not.

How is my performance in this class assessed?

What is the grading scale?*

93.0 - 100	A
90.0 - 92.99	A-
87.0 - 89.99	B+
83.0 - 86.99	B
80.0 - 82.99	B-
77.0 - 79.99	C+
73.0 - 76.99	C
70.0 - 72.99	C-
67.0 - 69.99	D+
63.0 - 66.99	D
60.0 - 62.99	D-
< 59.9	F

*no rounding, no curve

Your final course grade will be determined as follows:

35% - Participation: This has two components:

- **In-class activities (25%):** Each class will have an in-class activity component. These will be graded on completion and not correctness; however, it must be clear that a real attempt has been made for each question.
 - The grade for each assignment will be a 1 for complete, or 0 for incomplete (or not submitted). While we will take time in class to work on these activities, the goal is your understanding, thus you will have until the start of the next class period to complete the activity (and any optional bonus portion).
 - Late assignments will not be accepted. At the instructor's discretion, incomplete assignments may be sent back to students for corrections, due before the following class. If you are absent, you are still responsible for viewing the slides you missed, and completing this assignment and submitting on time.
- **Engagement (10%):** You are required to make substantial contributions while in class. During each class period, the TA will mark down who is present and actively engaged, and those will receive engagement credit. We hope that everyone receives full credit, but students using phones, or computers for non-class activities will receive a zero. We will put these on canvas the same day; if you believe an attendance mistake has been made, you must notify us prior to the next class.

Although ***makeup engagement points are not allowed***, I realize that you may need to occasionally miss class for a personal reason. For this reason, Canvas will ***drop your five lowest engagement grades***. Further excused absences will not be granted except in the case of official university business. You are responsible for completing the material you missed, still turning in the in-class activity prior to the next class. Should illness or other circumstances cause you to miss two or more class periods in a row, please contact [Student Outreach and Support](#) and they will contact me for accommodations.

Homework (20%): There will be periodic homework assignments to be completed fully outside of class. These will be submitted via canvas upload, and late work is not accepted. A small amount of extra credit may be offered on some homeworks. For students who turn in complete assignments on time, an opportunity will be available to make corrections and resubmit.

15% -Poster Project: Rather than a midterm exam, you will create a research poster based on a research paper you have read, and present in during a class period roughly mid-way through the semester. Further details will be available on Canvas.

30% -Final Project: This class will culminate with a collaborative research project, which you will complete in groups of two (one group of three is permitted if we have an odd number). During the semester we will have a few class periods dedicated to introducing you to a dataset. You and your group will pick a dataset, and craft a science question to explore, then submitting a project proposal as a group. Once your proposal is approved, you will work on your project (with several dedicated class periods, but expect to spend significant time outside of class). Each group member will submit their own, independent project report, in the form of a Research Note of the AAS (RNAAS; 1500 words, one figure or table), and you will collaboratively present your results during the last week of class. Further details

will be available on Canvas. ***All projects should include significant quantitative analysis including techniques learned in this class.*** Each group member must be responsible for a specific portion of the Python analysis, and the final grade will include an in-person walkthrough of your code.

What are other policies on exams, assignments, and other course structure?

Course Webpage: The course webpage on the Canvas system will be updated with course announcements, homework and reading assignments, and deadlines. It is your responsibility to check these on a regular basis. Please come to class prepared, having done any assigned reading and/or assignments. Also, please be prepared to participate in in-class discussions and activities, this is for your benefit.

Late work: Late work is not accepted. Makeup (or early) homework or projects are not offered. *However*, I understand that life events happen, so if you are unable to turn an item in on time, contact the **instructor in advance of the due date**. Note that being busy with other classes will not be considered a valid excuse. If you miss class for a sponsored University event, and you contact the instructor **in advance of the due date**, we can discuss accommodations.

Where can I find.. ?

Canvas will have:

1. Important announcements
2. Lecture slides
3. Assignments & modules
4. Syllabus
5. List of Learning Objectives
6. Gradebook

Canvas will be our main form of communication, so check it regularly and stay up to date on assignments and communications.

Course Conduct: Please silence cell phones before you enter the classroom. ***No texting or using your cell phone*** during class except for use in specified classroom activities. Please do not pack up or leave class early unless you have talked to me in advance, as a consideration to us and your fellow students. Students found to be using their computers for non-class activities will be a distraction to those around them, and will be asked to leave, and will not earn engagement credit for that day. ***Be respectful of others***, especially during in-class work times, and even if you disagree with them.

Students with Children: I recognize the difficulty of being a full time student with children. If you have children, or other family commitments, please come see me to discuss any modifications of the course policies which will maximize your success in this course.

Email: Email is recognized as an official mode of university correspondence; therefore you are responsible for reading your email for university and course-related information and announcements. Please check your email regularly and frequently (this includes Canvas announcements).

Use of AI:

The Use of AI: The creation of artificial intelligence tools for widespread use is an exciting innovation. These tools have both appropriate and inappropriate uses in classwork. The use of artificial intelligence tools (such as ChatGPT) in this class shall be permitted on a limited basis. Specifically:

- You may use ChatGBT (or any tool) to help find references. Be warned it is often inaccurate.

- You may **not** use ChatGBT (or any other tool) for writing - all words should be your own.
- AI tools can be useful coding aids. You may even find it can do your coding for you! However, the point of this class is to equip you with the knowledge to perform research — copying and pasting from an AI tool will not accomplish this.
- You may thus use any tool (from google, to stack exchange, to ChatGBT) to ***help find solutions*** to coding questions, but **all code should be your own**.
 - For example, if you run into the same problem again, you should remember how to code up the solution, rather than cutting and pasting from the internet.
- Important: **Any** time AI tools are used, they should be used with caution and proper citation, as the use of AI should be properly attributed. Using AI tools without my permission or authorization, or failing to properly cite AI even where permitted, shall constitute a violation of UT Austin's Institutional Rules on academic integrity.

If you are considering the use of AI tools but are unsure if you are allowed or the extent to which they may be utilized appropriately, please ask.

Classroom Safety and COVID-19

This is a science classroom and we are strongly recommending that we look to the science and follow the guidance of local public health officials and the CDC. There are several important ways YOU can help preserve the safety of our learning environment:

- **If you feel ill or have a positive COVID test please *DO NOT* come to class in person.** The activities and class notes will be posted online after class.
- If you experience cold-like symptoms or have close contact with someone who tested positive for COVID but have a negative COVID test, please do **wear a face covering** that covers your mouth and nose if you attend class as a courtesy to protect your peers.
- Visit protect.utexas.edu for more information.

Illness-related absences

We understand that illness often results in absences. This course has significant built-in flexibility, allowing you to miss up to five classes before your engagement grade is impacted. Should your illness extend more than a week, please complete a [Class Absence Notification Form](#) through Student Outreach and Support to provide illness documentation. Please do not share your medical information with me or the TA. Since course requirements and flexibility will vary between classes, schools and colleges, you are responsible for understanding expectations of your faculty members and any accommodations offered.

Frequently Asked Questions:

Do you record the in-class lectures?

No, this classroom is not set up to record. All class notes and materials will be posted online.

How do I succeed in this class?

The best way to succeed is to prepare and participate. Do any pre-class work ahead of time. When in class, buy in and ***participate***! After class, complete the in-class activity, and come to a help session to get help!

I'm sick, and can't attend class today, what do I do?

Stay home and get better (you can miss up to five classes, and still receive a full participation grade). Download the class slides after class and review them. Do the missed activities, in a group if you can find some classmates, or on your own, and submit prior to the next class. If you have to miss multiple consecutive classes, please contact Student Emergency Services, and they will let me know if they feel you should be allowed to make up for missed participation.

I need to leave class early? How do I make sure I don't lose participation?

Engagement is earned; you will not receive credit if you miss the bulk of a class period.

I forgot to turn in my assignment! Can I turn it in late?

No, late assignments or projects are not accepted. Don't let this be you, turn it in early!

Land Acknowledgement:

I would like to acknowledge that we are meeting on the Indigenous lands of Turtle Island, the ancestral name for what now is called North America.

Moreover, I would like to acknowledge the Alabama-Coushatta, Caddo, Carrizo/Comecrudo, Coahuil-tecan, Comanche, Kickapoo, Lipan Apache, Tonkawa and Ysleta Del Sur Pueblo, and all the American Indian and Indigenous Peoples and communities who have been or have become a part of these lands and territories in Texas. These people observed the night skies and marked the passing of the seasons here long before the founding of Texas.

Course Calendar: *Schedule of topics subject to change

Class	Dates	Topics*	Project Due Dates
1	Aug 26 (T)	Introduction to AST376R, Get to know your Computer, UNIX basics	
2	Aug 28 (Th)	What is Research, Department research events and opportunities, NASA ADS	
3	Sept 2 (T)	FITS images, Coordinate systems, SAOImage ds9	
4	Sept 4 (Th)	Ds9 Continued	
5	Sept 9 (T)	Introduction to Python	
6	Sept 11 (Th)	How to make a poster, discussion of poster project	
7	Sept 16 (T)	Python I/O, Intro to tables	

Class	Dates	Topics*	Project Due Dates
8	Sept 18 (Th)	Python I/O, PANDAS	Poster paper choice due 11:59p
9	Sept 23 (T)	Plotting in Python	
10	Sept 27 (Th)	Class Project I	
11	Sept 30 (T)	How to write and manage citations: Latex, overleaf, bibtex, bibdesk, zotero	
12	Oct 2 (Th)	Class Project II	
13	Oct 7 (T)	Class Project Brainstorming	Posters due 11:59p
14	Oct 9 (Th)	Poster Presentations	Project Abstract due 11:59p
15	Oct 14 (T)	Python Analysis I	
16	Oct 16 (Th)	How to Write a Proposal	
17	Oct 21 (T)	Project Work time	Project Proposal due 11.59p
18	Oct 23 (Th)	Python Analysis II: Spectroscopy	
19	Oct 28 (T)	Databases & Astroquery	
20	Oct 30 (Th)	Python Analysis III: MCMC	
21	Nov 4 (T)	Project work time	
22	Nov 6 (Th)	Spectral Cubes	
23	Nov 11 (T)	CVs and Networking	Draft of project report due 2pm
24	Nov 13 (Th)	Project report peer review, project work time	
25	Nov 18 (T)	Careers in astronomy, project work time	
26	Nov 20 (Th)	Project work time	
27	Dec 2 (T)	Project presentations	Final report and presentation slides due 2p
28	Dec 4 (Th)	Project presentations	

University Resources:

Academic accommodations (D&A): This class respects and welcomes students of all backgrounds, identities, and abilities. If there are circumstances that make our learning environment and activities difficult, or if you have medical information that you need to share with me, please let me know. I am committed to creating an effective learning environment for all students, but I can only do so if you discuss your needs with me as early as possible. I promise to maintain the confidentiality of these discussions. Any student with a documented disability who requires academic accommodations should contact Disability & Access at 512-471-6259 (voice) or 512-410-6644 (Video Phone) as soon as possible to request an official letter outlining authorized accommodations (email: access@austin.utexas.edu). For more information, visit <https://disability.utexas.edu/>. *I am also happy to meet to discuss more.*

Counseling and Mental Health Center: Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress. All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. <http://www.cmhc.utexas.edu/individualcounseling.html>

The Sanger Learning Center: Did you know that more than one-third of UT undergraduate students use the Sanger Learning Center each year to improve their academic performance? All students are welcome to take advantage of Sanger Center's classes and workshops, private learning specialist appointments, peer academic coaching, and tutoring for more than 70 courses in 15 different subject areas (including Astronomy). For more information, please visit <http://www.utexas.edu/ugs/slc> or call 512-471-3614 (JES A332).

University and Course Policies:

Academic integrity: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. Students who violate University rules on scholastic dishonesty are subject to disciplinary penalties. Ethical conduct is expected at all times. For example, violating our course AI policy, or copying code from others is unethical. Incidences of academic dishonesty will be reported to Student Conduct and Academic Integrity. You are responsible for understanding UT's Academic Honesty and the University Honor Code which can be found at the following web address: <https://deanofstudents.utexas.edu/conduct/standardsofconduct.php>

- **Sharing of Course Materials is Prohibited:** No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated

with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Plagiarism: As a research university, the University of Texas at Austin takes plagiarism very seriously. Do not risk getting involved in a plagiarism infraction - the consequences simply aren't worth it. Always cite your sources, and when in doubt consult a professor or librarian.

Personal or Family Emergencies: If you experience a personal or family emergency (death in the family, protracted sickness, serious mental health issues) that prevents you from attending an exam or forces you to miss multiple days of class, you should contact Student Outreach and Support in the Office of the Dean of Students <https://deanofstudents.utexas.edu/sos/index.php>. They will work with you to communicate with your professors and let them know of your situation.

Religious Days: A student who is absent from a class or examination for the observance of a religious holy day will be permitted to make up the missed work, if notice is given at least fourteen days prior to such an absence.

Title IX: Beginning January 1, 2020, Texas [Senate Bill 212](#) requires all employees of Texas universities, including faculty, report any information to the [Title IX Office](#) regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.